



An offline-first Android app that checks your medicines for dangerous interactions, reminds you when doses are due, and explains the risk in plain language.

Healthtech & Wellness

Works offline

₹0 to run

No account

The problem

Roughly half of all long-term medicine is not taken as prescribed, and taking five or more medicines — normal for an older adult — roughly doubles the risk of a harmful reaction. In India much of that medicine is bought over the counter, so no single doctor or pharmacist ever sees the full list.

Reminder apps tell you *when* to take something but never whether it is safe. Interaction checkers answer that, but they need the internet, are written in clinical English, and are built for pharmacists rather than patients. MediBloom does both, offline, in plain words.

What it does

Interaction checking

Every pair of your medicines is checked the moment you add one, against 134 bundled rules covering 130 generic drugs. Indian brand names resolve automatically — type “Shelcal”, not “calcium carbonate”.

Smart Swap

Where a safer alternative exists, the app names it. It always swaps the painkiller, never the medicine keeping you alive.

Reminders

Mark taken / Snooze / Skip sit on the notification itself. Exact alarms, survive a reboot.

Prescription scanning

Photo or PDF, read on the phone by Google ML Kit. It shows what it saw and asks before saving anything.

Caregiver alerts

If a dose stays unmarked past your chosen limit, a family member is emailed automatically.

Ask MediBloom

A chat that answers from your own dose log. “What did I have today?” is answered instantly on the phone; open-ended questions can go to a free AI model.

Wellness dashboard

Streak, adherence percentage, a tappable four-week calendar, week-on-week trend and a per-medicine breakdown — all from your own history.

Accessibility

Larger-text mode, voice read-aloud, touch targets of at least 44px, plain language everywhere, and a time picker built from big + and – buttons.

How it works

Everything that matters runs on the phone. The interaction rules are compiled into the app, the data lives in a SQLite database on the device, reminders are handed to Android’s alarm system, and the OCR runs locally. There is no server anywhere.

The assistant is split deliberately. Questions that are a direct read of your own data are answered by the rule engine — instant, exact, offline. Only open-ended questions go to an AI model, and if that is slow, rate-limited or unreachable, the rules answer instead and the app says which one replied.

Technology used

App	React Native 0.86 · Expo SDK 57 · TypeScript (strict) · React Navigation 7
Database	expo-sqlite — on the phone, the only place data lives
Reminders	expo-notifications — daily triggers on Android AlarmManager
OCR	Google ML Kit via @react-native-ml-kit/text-recognition — fully offline
PDF reading	Custom Expo module in Kotlin, wrapping Android's own PdfRenderer
Secrets	expo-secure-store — Android Keystore
Fonts & icons	DM Serif Display + Nunito · all icons hand-drawn in react-native-svg
Build	Gradle signed release APK · Jest (221 tests) · adb UI testing on device

External services (all optional, all free)

Groq	Optional AI for open-ended questions	openai/gpt-oss-120b, free tier
Google AI Studio / OpenRouter	Alternative AI providers	free tier
Email JS	Sending caregiver alerts automatically	free, 200 emails a month
Google Apps Script	Alternative: sends via the user's own Gmail	free

None of these are needed for the app to work. With no AI key the assistant answers from its own rules; with no email relay the caregiver alert arrives as a notification with the email already written, one tap from sending.

Key numbers

134 interaction rules	130 generic drugs	398 brand names	221 tests passing	12 screens	₹0 cost per user
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Demo in six steps

1. Open **Settings** → **Load demo data**. This loads Lakshmi, 68, from Chennai.
2. **Home** — point at the red banner: Warfarin + Meftal. The insight below it is generated from her real dose log.
3. **Tap the banner** → **Interactions**. Read the severe card, then the **Meftal** → **Acetaminophen** swap.
4. **Medicines** → **Upload a prescription** → **Try our sample**. Six medicines parsed; one deliberate misread flagged “Please check this”.
5. **Assistant** — type “what did I have today” (instant, offline), then “my knee is hurting what can I take” (goes to the AI).
6. **Settings** → **Send a test reminder**. Pull down the shade, tap Mark taken, return Home — it is ticked off.

If something goes wrong on stage

If the chat is slow, that is the fallback working — say so. If a notification does not fire, use **Settings** → **Send a test reminder**. If OCR fails on a photo, use **Try our sample prescription**, which runs the identical parser. If nothing works at all, open **Interactions** — it needs no network, no permissions and no notifications.

Questions you will be asked

How does it work without internet?

Everything is inside the app. The interaction rules are compiled into the APK, the data is in a SQLite database on the phone, reminders are Android alarms, and the OCR runs locally. There is no server to be offline from.

Where does the drug data come from?

We wrote it — 134 well-established, textbook-level interactions. It is not a certified drug database, and it would need a pharmacist's review before real patients use it.

Did you train an AI model?

No. A model that invents a drug interaction is more dangerous than no answer at all. Ours is a deterministic lookup, so every warning traces back to a rule we can show you.

So where is the AI used?

Only for open-ended questions the rules cannot cover, like "my knee is hurting, what can I take". Factual questions about your own log are answered on the phone, instantly. If the AI fails or you are offline, the rules answer instead and the app says so.

Can the AI say something dangerous?

It only sees the user's own data and is instructed to answer from that alone. It is never the source of an interaction warning, and any medicine name it suggests is discarded unless it matches our own reference.

Where is the user's data stored?

In SQLite on the phone, and nowhere else. No account, no analytics, no tracking. The only things that can leave are an AI question or a caregiver email — both optional, both labelled.

What does it cost to run?

Nothing, per user, forever. No servers, no database, no API costs in the default path. The optional AI and email both use free tiers.

How does it scale?

There is nothing to scale. Every install is independent — a million users is a million phones doing their own work, and our cost stays at zero.

How would an elderly person use it?

That shaped the design. Big buttons, larger-text mode, voice read-aloud, brand names instead of chemical names, plain words instead of jargon, and no login at all. Caregiver alerts exist because realistically a son or daughter helps.

How do you know it works?

221 automated tests over the logic, plus we drove the real signed APK on a device and checked every screen. The logs are clean — no crashes, no errors.

Why Android and not a website?

A website cannot set an alarm that fires when the phone is locked, cannot work offline reliably, and cannot read a prescription with the camera. Reminders are the core of the product.

What makes it different from existing apps?

Reminder apps do not check interactions. Interaction checkers need the internet and are written for pharmacists. Nobody does both, offline, for the person actually at risk.

What we would say before being asked

- The interaction dataset is hand-built, not certified. It needs pharmacist review before real patients.
- Handwriting OCR is unreliable — so the app shows what it read and asks, rather than guessing.
- Some Android manufacturers throttle background alarms unless the app is exempted from battery optimisation.
- Cloud assist sends a short summary off the phone. It is optional and clearly labelled.
- The demo build ships API keys inside the APK so it works with no setup — revoke them before sharing widely.

MediBloom provides general medication-safety support. It does not diagnose, prescribe, or replace a doctor or pharmacist. In an emergency, contact local emergency services. The patients, hospitals and prescriptions used in the demo are fictional.